PART I / SELECTION TO REACTION

When do I use this reagent?

Start with the reagent in the left column. Follow its reaction number for the full combination of reactants, catalyst and heating.

Reagent in Bond Lab	When to choose it	Reaction
H_2	Saturate an alkene double bond using a metal catalyst.	**04**
Cl_2 or Br_2	Replace one H in methane using UV, or add two halogen atoms across an alkene in organic solvent.	**02**, **05**
I_2	Alkene addition in organic solvent is a game option. See the iodine note on page 3.	**05**
HCl, HBr or HI	Add HX across an alkene, or replace OH in an alcohol with the corresponding halogen. HCl also protonates carboxylates and amines.	**06**, **10**, **28**, **29**
PCl_3 or PBr_3	Replace an alcohol's OH with Cl or Br. The selector groups each reagent with its corresponding hydrogen halide.	**10**
NaOH(aq)	Heat for haloalkane substitution or ester/amide hydrolysis. Use without heat to deprotonate a carboxylic acid or ammonium ion.	**08**, **22**, **23**, **25**, **26**, **27**, **30**
$K_2Cr_2O_7$(aq) / H^+(aq)	Oxidise a primary or secondary alcohol, or an aldehyde. For a primary alcohol the heating method determines which product you collect.	**12–15**
$NaBH_4$(aq)	Reduce an aldehyde or ketone to an alcohol.	**16**, **17**
$LiAlH_4$ in dry ether, then H^+(aq)	Reduce an aldehyde, ketone or carboxylic acid. Choose the combined two-stage option.	**16–18**
Concentrated H_2SO_4	Dehydrate an alcohol, or catalyse esterification of a carboxylic acid with an alcohol.	**11**, **19**
Dilute H^+(aq)	Hydrolyse an ester or amide in acid, or protonate a carboxylate or amine.	**21**, **24**, **28**, **29**
NH_3	Form an unsubstituted amide from a carboxylic acid, via the ammonium salt.	**20**
Excess O_2	Completely burn an alkane to carbon dioxide and water.	**03**

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PART I / SELECTION TO REACTION

Catalysts, heat and final steps

Choose all the selections for one method. Where alternatives are given, choose one complete method. Add the organic partner only in the step where it reacts.

Catalyst selector	When to choose it
Pt / Ni / Pd	Hydrogenate an alkene with H_2 (**04**). Any one of the three metals catalyses the reaction.
Al_2O_3	Dehydrate an alcohol with heat (**11**). This is an alternative to concentrated sulfuric acid.
Silica / alumina	Crack a longer-chain alkane with strong heat (**01**).
None	All other methods, including acid-catalysed methods where the acid is selected in the reagent list.

Heating

Heating selector	When to choose it
None	Methane substitution (with UV); alkene additions **04–06**; alcohol substitution **10**; reductions **16–18**; acid–base reactions **27–30**.
Heat	Required for dehydration (**11**) and amide formation (**20**). Also accepted wherever pages 3–5 say *heat or reflux*.
Heat under reflux	Condense vapour back into the flask while heating. Required for primary alcohol → acid (**13**). Also accepted for **08**, **14**, **15**, **19** and **21–26**.
Heat and distil	For primary alcohol → aldehyde (**12**). Remove the aldehyde as it forms to limit further oxidation.
Strong heat	Crack a longer-chain alkane with silica / alumina (**01**).
Ignition	Start complete combustion with excess oxygen (**03**).

Light, work-up and organic partners

Selection	When to choose it
UV light	Initiate methane chlorination or bromination (**02**). Alkene halogen addition (**05**) uses organic solvent instead.
Then $H^+(aq)$	After alkaline ester or amide hydrolysis, add excess acid to obtain the free acid (**23**, **26**). For $LiAlH_4$ reductions (**16–18**), the combined reagent already includes this final step. Keep the reduction dry until work-up.
Methanol, ethanol or propan-1-ol	Choose one alcohol for esterification (**19**). Methanol gives a methyl ester, ethanol an ethyl ester and propan-1-ol a propyl ester.

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PART II / REACTION TO SELECTIONS

Hydrocarbons and haloalkanes

Find the change you want in the middle column. The last column gives the complete game method. Leave unspecified options at their defaults.

ID	Starting material and products	What to select
01	Longer-chain alkane → smaller alkane + alkene (cracking)	Silica / alumina catalyst + strong heat. Use the supplied product to work out the carbon count of its partner.
02	Methane → chloromethane or bromomethane + HX	Cl_2 or Br_2, respectively, + UV light. HX is HCl for chlorination and HBr for bromination.
03	Alkane → carbon dioxide + water (complete combustion)	Excess O_2 + ignition.
04	Alkene → alkane (hydrogenation)	H_2 + Pt / Ni / Pd catalyst.
05	Alkene → vicinal dihaloalkane	Select Cl_2, Br_2 or I_2 in organic solvent. One halogen atom adds to each carbon of the former double bond.
06	Alkene → monohaloalkane	HCl, HBr or HI. For an unsymmetrical terminal alkene, H adds to the double-bond carbon already bearing more H atoms.
08	Haloalkane → alcohol (substitution)	NaOH(aq) + heat or reflux. Replace the halogen with OH at the same carbon.

Haloalkane hydrolysis

Starting with bromoethane, aqueous NaOH and heat give ethanol (**08**). The OH group replaces Br on the same carbon. The same substitution method applies to the other supported haloalkanes.

Halogen choice

For a standard example of addition across an alkene, use bromine. Bond Lab also accepts iodine in organic solvent; ordinary alkene iodination is generally thermodynamically unfavourable. See the chemistry reference on page 6.

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PART II / REACTION TO SELECTIONS

Alcohols and carbonyl compounds

Identify the starting functional group and the oxidation level needed at the end. The same oxidising agent can give different products.

ID	Starting material and products	What to select
10	Alcohol → chloro-, bromo- or iodoalkane	For Cl, use HCl or PCl_3. For Br, use HBr or PBr_3. For I, use HI. The selector groups the alternatives for Cl and Br.
11	Alcohol → alkene + water (dehydration)	Concentrated H_2SO_4 + heat, or Al_2O_3 catalyst + heat. The game uses terminal alcohols or unsubstituted cycloalkanols with an adjacent C–H.
12	Primary alcohol → aldehyde	Acidified $K_2Cr_2O_7$(aq) + heat and distil. Collect the aldehyde as it forms.
13	Primary alcohol → carboxylic acid	Acidified $K_2Cr_2O_7$(aq) + heat under reflux. Use sufficient oxidant for full oxidation.
14	Secondary alcohol → ketone	Acidified $K_2Cr_2O_7$(aq) + heat or reflux.
15	Aldehyde → carboxylic acid	Acidified $K_2Cr_2O_7$(aq) + heat or reflux.
16	Aldehyde → primary alcohol	$NaBH_4$(aq), or $LiAlH_4$ in dry ether followed by H^+(aq).
17	Ketone → secondary alcohol	$NaBH_4$(aq), or $LiAlH_4$ in dry ether followed by H^+(aq).
18	Carboxylic acid → primary alcohol	$LiAlH_4$ in dry ether followed by H^+(aq). Choose the combined reagent.

Choosing the apparatus

Ethanol gives ethanal when the aldehyde is distilled out (**12**). Keeping the mixture under reflux allows oxidation to continue to ethanoic acid (**13**). Starting from ethanal itself, choose **15**.

Apply the same methods to rings

Cyclohexanol is a secondary alcohol. Oxidation gives cyclohexanone (**14**), and reduction of cyclohexanone returns cyclohexanol (**17**). Dehydration of cyclohexanol gives cyclohexene and water (**11**). The six-carbon ring remains intact.

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PART II / REACTION TO SELECTIONS

Esters and amides

For hydrolysis, first identify the acid-derived fragment and the alcohol- or nitrogen-derived fragment. The final conditions determine their charges.

ID	Starting material and products	What to select
19	Carboxylic acid + alcohol → ester + water	Methanol, ethanol or propan-1-ol + concentrated H_2SO_4 + heat or reflux.
20	Carboxylic acid + ammonia → unsubstituted amide + water	NH_3 + heat. First form the ammonium carboxylate, then heat it to form the amide.
21	Ester → carboxylic acid + alcohol	Dilute H^+(aq) + heat or reflux.
22	Ester → carboxylate + alcohol	NaOH(aq) + heat or reflux.
23	Ester → carboxylic acid + alcohol, via alkaline hydrolysis	NaOH(aq) + heat or reflux, **then** H^+(aq).
24	Amide → carboxylic acid + ammonium species	Dilute H^+(aq) + heat or reflux.
25	Amide → carboxylate + ammonia or amine	NaOH(aq) + heat or reflux.
26	Amide → carboxylic acid + ammonium species, via alkaline hydrolysis	NaOH(aq) + heat or reflux, **then** excess H^+(aq). The acid protonates both fragments.

Worked example: *N*-propylbutanamide

Break the bond between the carbonyl carbon and nitrogen. The four-carbon fragment comes from butanoic acid. The three-carbon propyl group stays attached to nitrogen.

Final conditions	Acid-derived product	Nitrogen product
Acidic (24, 26)	Butanoic acid $CH_3CH_2CH_2COOH$	Propylammonium $CH_3CH_2CH_2NH_3^+$
Alkaline (25)	Butanoate $CH_3CH_2CH_2COO^-$	Propan-1-amine $CH_3CH_2CH_2NH_2$

If one product is shown, draw its partner. For an unsubstituted amide ($RCONH_2$), the nitrogen product is NH_4^+ after acid treatment or NH_3 under alkaline conditions.

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PART II / REACTION TO SELECTIONS

Acid–base changes and synthesis

If the skeleton and functional group connectivity stay the same but the charge changes, look for a proton transfer. These four steps require no heating.

ID	Starting material and products	What to select
27	Carboxylic acid → carboxylate ion + water	NaOH(aq). The acid loses the H from its OH group.
28	Carboxylate ion → carboxylic acid	Dilute H^+(aq) or HCl. Protonate the negatively charged oxygen.
29	Amine → alkylammonium ion	Dilute H^+(aq) or HCl. Nitrogen gains a proton and a positive charge.
30	Alkylammonium ion → amine + water	NaOH(aq). Hydroxide removes a proton from nitrogen.

Worked route: 1-bromobutane to butanoate

This challenge requires exactly three steps, the minimum using the supported reactions. All four carbons stay together. First replace Br with OH, oxidise the terminal alcohol fully, then remove the acid proton.

Step	Conversion	One accepted method
1	1-Bromobutane → butan-1-ol	NaOH(aq), heat (08).
2	Butan-1-ol → butanoic acid	Acidified $K_2Cr_2O_7$(aq), heat under reflux (13).
3	Butanoic acid → butanoate ion	NaOH(aq), no heating (27).

Scope and references

Reaction numbers retain their original labels. Entries 07 (alkene hydration) and 09 (haloalkane elimination) have been removed from this guide and the game.

Methods and selectable alternatives follow Bond Lab's reaction generator 2026.7. The game uses straight-chain and selected five- and six-membered ring examples. The tables name distinct products and omit spectator ions and balancing coefficients.

Hong Kong Education Bureau: Chemistry notes for teachers (curriculum context).

OpenStax Organic Chemistry, 8.2: Halogenation of alkenes (iodine comparison).

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